

Name _____



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

Class	Reg Number
17	

Calculator Model / No.

Chemistry

8873/01

Paper 1

21 September 2018
1 hour

Additional Material: *Data Booklet*
 OMS

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces provided at the top of this page.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the use of OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of OMR Answer Sheet

Ensure you have written your name, class register number and class on the OMR Answer Sheet.

Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **OMR sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

This document consists of **12** printed pages (including this cover page)

Answer all questions.

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 Zinc ethanoate, $(\text{CH}_3\text{CO}_2)_2\text{Zn}$ ($M_r = 183.4$) may be taken as a dietary supplement to prevent zinc deficiency.

What is the total number of ions present in a 5 cm³ solution of aqueous zinc ethanoate given that the solution has a concentration of 10.64 g dm⁻³?

- A** 1.7×10^{19} **B** 5.2×10^{20} **C** 3.5×10^{22} **D** 1.0×10^{23}

- 2** Gases given off during volcanic eruptions include H_2S and CS_2 .

A 40 cm³ gaseous sample of H₂S and CS₂, present in a 3 : 1 ratio respectively, was analysed by combustion using 100 cm³ of oxygen. After measuring the volume of gas remaining, the product was treated with an excess of aqueous sodium hydroxide and the volume of gas measured again.

Any sulfur present is converted to SO_2 after combustion.

Given that all volume measurements were made under room conditions, what were the measured volumes?

	volume of gaseous mixture after burning / cm ³	volume of gaseous mixture after adding NaOH (aq) / cm ³
A	60	0
B	60	50
C	85	25
D	85	75

- 3** Na_3NO_4 is very sensitive to water vapour and carbon dioxide in the air and it reacts with both of them to produce equimolar amounts of NaNO_3 , NaHCO_3 and compound **E**.

What is compound **E**?

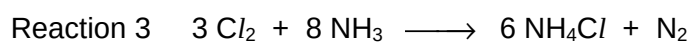
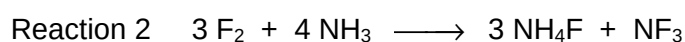
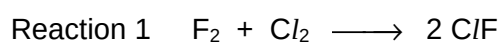
- A** NaH
B Na₂O
C NaOH
D NaNO₂

4 Which of the following contains more than one type of chemical bond?

- 1 barium carbonate
- 2 graphene
- 3 water

- | | | | |
|----------|---------------------|----------|---------------------|
| A | 1, 2 and 3 | B | 1 and 2 only |
| C | 1 and 3 only | D | 1 only |

5 Consider the following reactions:



Which of the following correctly shows the increasing order of reducing strength of the three species, Cl_2 , F_2 and NH_3 ?

- | | | | |
|----------|--|----------|--|
| A | $\text{F}_2 < \text{Cl}_2 < \text{NH}_3$ | B | $\text{Cl}_2 < \text{F}_2 < \text{NH}_3$ |
| C | $\text{NH}_3 < \text{F}_2 < \text{Cl}_2$ | D | $\text{NH}_3 < \text{Cl}_2 < \text{F}_2$ |

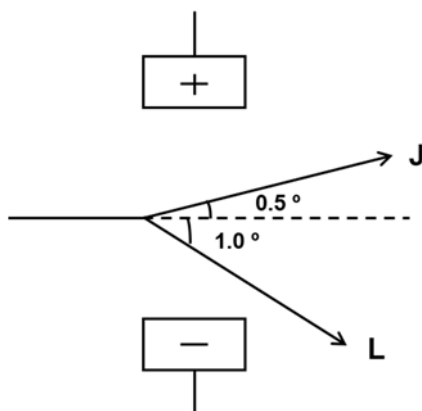
6 Which of the following statements about Al is correct?

- 1 It has high electrical conductivity due to the ability of its ions to carry the current.
- 2 It has high melting point due to strong attraction between the delocalised electrons and residual cations.
- 3 It has stronger metallic bonding than Na.

- | | | | |
|----------|-------------------|----------|---------------------|
| A | 1, 2 and 3 | B | 1 and 2 only |
| C | 1 only | D | 2 and 3 only |

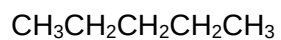
- 7 In the presence of an electric field, which of the following pair of ions will demonstrate deflection pathways as shown in the diagram below?

In this particular experimental setup, protons are deflected through an angle of 7.0° .

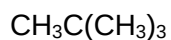


	J	L
A	$^{14}\text{N}^-$	$^7\text{Li}^+$
B	$^{18}\text{O}^-$	$^{27}\text{Al}^{3+}$
C	$^{16}\text{O}^+$	$^{32}\text{S}^{2-}$
D	$^{28}\text{Si}^{2+}$	$^{28}\text{Si}^{4-}$

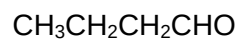
- 8 A comparison is made of four different organic compounds with the same molar mass.



W



X



Y



Z

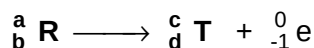
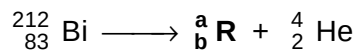
Which of the following shows the boiling points of these compounds in the correct sequence?

	highest			lowest
A	W	X	Y	Z
B	Y	Z	W	X
C	Z	Y	W	X
D	Z	Y	X	W

- 9 Use of the Data Booklet is relevant to this question.

Bismuth-212 is an unstable isotope of bismuth with a half-life of around 60 min.

The radioactive decay of bismuth-212 follows the following scheme, giving rise to two particles, **R** and **T**:

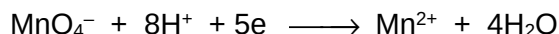


Which are the likely formulae of the chlorides formed by particles, **R** and **T**?

	chloride of R	chloride of T
A	RCl₂	TCl₂
B	RCl₂	TCl₃
C	RCl₃	TCl₂
D	RCl₃	TCl₃

- 10 A 25 cm³ of solution containing 0.010 mol dm⁻³ of an ion **G**⁴⁺ (aq) was reduced by excess zinc. After the unreacted zinc was removed, the resultant solution required 20.0 cm³ of 0.0050 mol dm⁻³ potassium manganate(VII) to oxidise **G** back to its original +4 oxidation state.

MnO₄⁻ is reduced according to the equation:

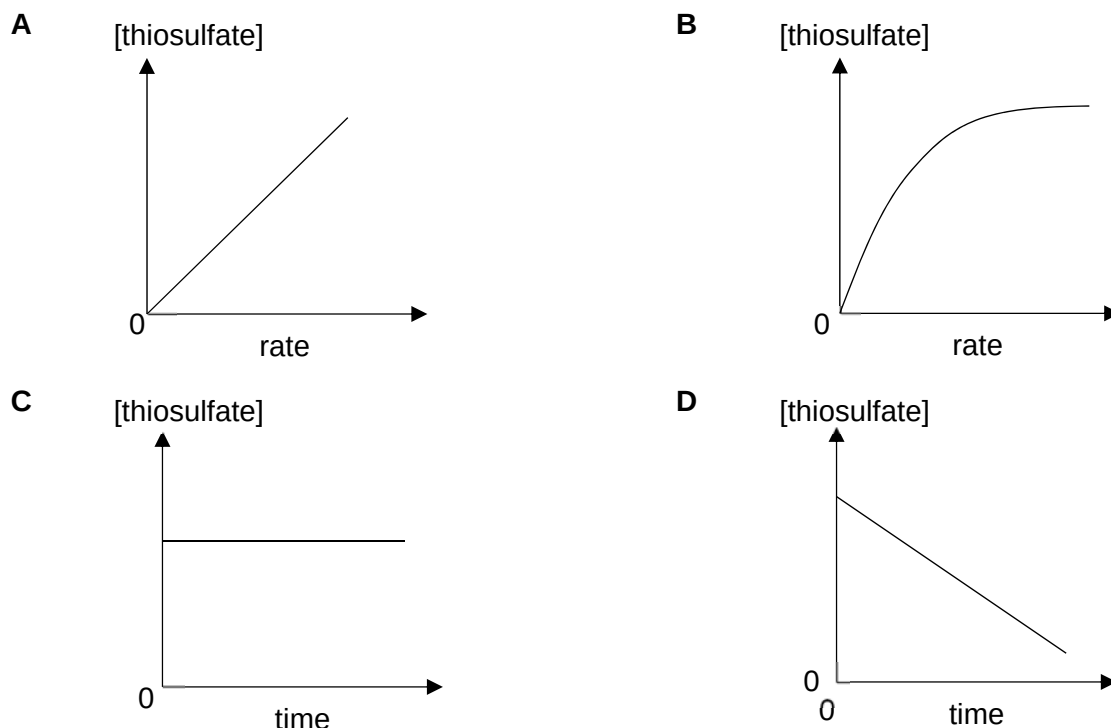


To what oxidation state was **G**⁴⁺ reduced by zinc?

- A** +3 **B** +2 **C** +1 **D** 0
- 11 The hydrolysis of the anticancer drug *cis-platin* in water follows a first-order kinetics with a rate constant of 0.09 h⁻¹ at 25 °C.
- How long will it take for the concentration of a freshly prepared aqueous solution of *cis-platin* to decrease to 18% of its original concentration?
- A** 7.7 h **B** 15.4 h
C 19.0 h **D** 42.8 h

- 12 In the reaction between aqueous sodium thiosulfate and dilute acid, the reaction is found to be first order with respect to thiosulfate.

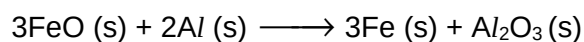
Which graph represents the experimental results?



- 13 Which statement **best** defines the term *bond energy* for HF?

- A Energy released when one mole of HF is formed from its gaseous ions.
- B Energy released when one mole of HF is formed from its gaseous atoms.
- C Energy required when one molecule of HF is broken into its gaseous atoms.
- D Energy required when one mole of HF is broken into its gaseous atoms.

- 14 The values for the standard enthalpy changes of formation of iron(II) oxide, FeO (s), and aluminium oxide, Al₂O₃ (s) are -266 kJ mol^{-1} and $-1676 \text{ kJ mol}^{-1}$ respectively.



What is the standard enthalpy change for the above reaction?

- | | |
|-------------------------------|-------------------------------|
| A $+878 \text{ kJ mol}^{-1}$ | B -878 kJ mol^{-1} |
| C $+1410 \text{ kJ mol}^{-1}$ | D $-1410 \text{ kJ mol}^{-1}$ |

15 Which of the following enthalpy changes is always exothermic?

- 1 standard enthalpy change of formation
- 2 standard enthalpy change of combustion
- 3 standard enthalpy change of neutralisation
- 4 standard enthalpy change of reaction

A 1, 2, 3 and 4

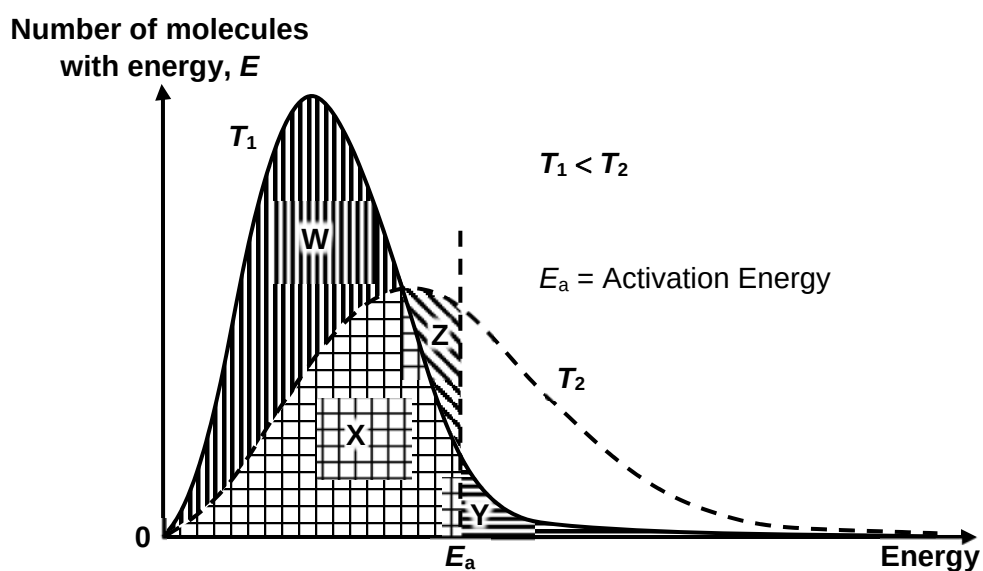
B 1 and 2 only

C 2 and 4 only

D 2 and 3 only

16 The distribution of the fraction of molecules with energy for two temperatures, T_1 and T_2 are shown in the diagram below.

W, X, Y and Z refer to the different shaded areas under the two graphs.



Which of the following expressions gives the **increase** in fraction of molecules having at least the activation energy at higher temperature, T_2 ?

A $\frac{Y}{X + Y + Z}$

B $\frac{Z}{X + Y + Z}$

C $\frac{X - Z}{W + X + Y}$

D $\frac{W - Z}{W + X + Y}$

17 Which of the following statements about a catalyst is correct?

- 1 A catalyst increases both the rate of the forward and backward reaction.
- 2 A catalyst increases the rate of reaction through affecting the rate constant.
- 3 A catalyst increases the rate of reaction through decreasing the enthalpy change of the reaction, ΔH .
- 4 A catalyst increases the rate of reaction through increasing the average kinetic energy of the reactant particles.

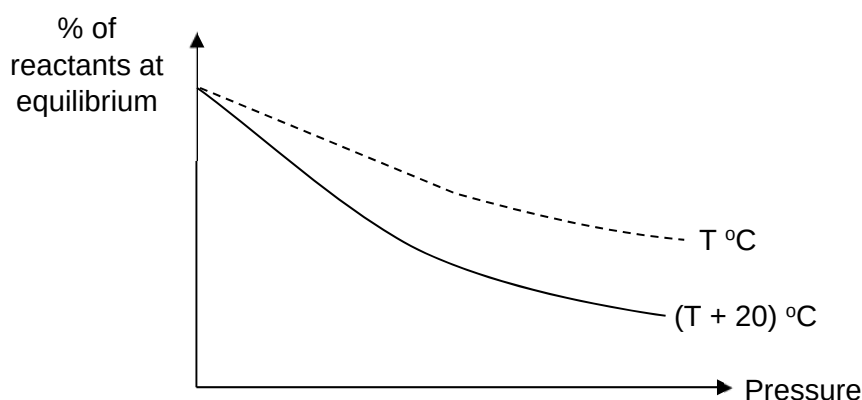
A 1, 2 and 4 only

B 1 and 2 only

C 2 and 3 only

D 2 and 4 only

18 The graphs below show the variation of the percentage of gaseous reactants present at equilibrium, with temperature and pressure.



Which of the following systems could the graphs represent?

- | | | |
|---|---|--|
| A | $2\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{N}_2\text{O}(\text{g})$ | $\Delta H = +82 \text{ kJ mol}^{-1}$ |
| B | $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g})$ | $\Delta H = +173 \text{ kJ mol}^{-1}$ |
| C | $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ | $\Delta H = -92 \text{ kJ mol}^{-1}$ |
| D | $3\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g}) \rightleftharpoons 2\text{N}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ | $\Delta H = -1248 \text{ kJ mol}^{-1}$ |

- 19** Platinum and rhodium nanoparticles are used in catalytic converters of motor cars.

Which reaction in the catalytic converter does **not** remove hazardous and polluting gases from the exhaust fumes?

These equations are qualitative and unbalanced.

[HC = unburnt hydrocarbon; NO_x = oxides of nitrogen]

- A** $\text{HC} + \text{NO}_x \longrightarrow \text{H}_2\text{O} + \text{CO} + \text{N}_2$
- B** $\text{CO} + \text{NO}_x \longrightarrow \text{CO}_2 + \text{N}_2$
- C** $\text{HC} + \text{NO}_x \longrightarrow \text{H}_2\text{O} + \text{CO}_2 + \text{N}_2$
- D** $\text{CO} + \text{O}_2 \longrightarrow \text{CO}_2$

- 20** What is the volume of deionised water that needs to be added to a 10 cm³ of 0.0100 mol dm⁻³ calcium hydroxide to have a resultant solution with pH of 11.5?

- | | | | |
|----------|----------------------|----------|----------------------|
| A | 21.6 cm ³ | B | 31.6 cm ³ |
| C | 53.2 cm ³ | D | 63.2 cm ³ |

- 21** The table shows some data of two acid–base indicators.

indicator	approximate pH range of colour change	colour change	
		acid	alkali
bromophenol–blue	3.0 – 4.6	yellow	purple
phenol–red	6.8 – 8.5	yellow	red

Which conclusion can be drawn about a solution when it turns bromophenol–blue purple and phenol–red yellow?

- A** The solution is strongly acidic.
B The solution is weakly acidic.
C The solution is strongly alkaline.
D The solution is weakly alkaline.

- 22** Which of the following is correctly arranged in order of **decreasing** values?

- A** atomic radius of P, S, Cl
B melting point of P, S, Cl
C first ionisation energy of Na, Mg, Al
D electrical conductivity of Na, Mg, Al

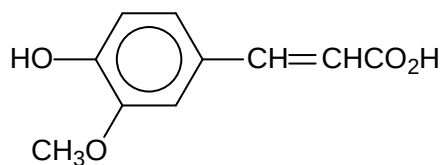
- 23 The properties of the oxides of four elements in Period 3, **E**, **F**, **G** and **H** are given below.

- 1 The oxide of **E** is amphoteric.
- 2 The oxide of **F** dissolves in water to form a strongly alkaline solution.
- 3 The oxide of **G** reacts with dilute sodium hydroxide at room temperature.
- 4 The oxide of **H** is insoluble in water but is soluble in concentrated sodium hydroxide.

Which of the following shows the correct sequence of the four elements in order of increasing proton number?

- A **F, E, G, H**
- B **F, E, H, G**
- C **G, F, E, H**
- D **G, F, H, E**
- 24 Which of the following statements explains the trend in thermal stability of hydrogen halides down the group?
- A Hydrogen halide bond becomes less polar down the group.
- B Electron cloud size of hydrogen halide molecules increases down the group.
- C Bond energy of hydrogen halide decreases more than the bond energy of the halogens.
- D Less energy is required to break the permanent dipole – permanent dipole attractive forces in the hydrogen halides down the group.
- 25 How many isomers (including both constitutional and cis–trans isomers) are possible for C_4H_7Cl ?
- A 9 B 10 C 11 D 12

- 26 Ferulic acid is an antioxidant that occurs widely in plants.

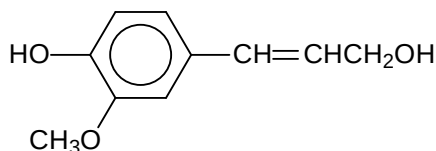


ferulic acid

Which of the following statements is true about ferulic acid?

- 1 It decolourises aqueous bromine.
- 2 It is not very soluble in water but dissolves in aqueous NaOH.

3

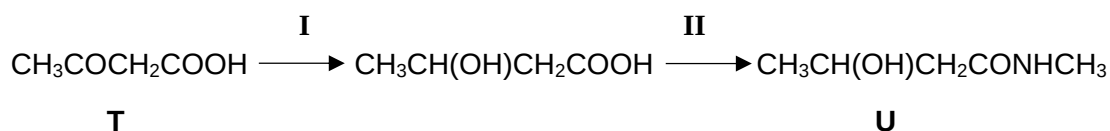


It can be prepared from

using hot acidified $K_2Cr_2O_7$.

- | | | | |
|----------|-------------------|----------|---------------------|
| A | 1, 2 and 3 | B | 1 and 2 only |
| C | 1 only | D | 2 and 3 only |

- 27 Compound **T** can be converted to a compound **U** as shown below.



Which of the following statements is **incorrect** with regards to the given reaction scheme?

- A** Stage **I** may involve the use of sodium borohydride.
- B** Stage **I** may involve the use of lithium aluminium hydride.
- C** Stage **II** may involve the use of DCC and methylamine.
- D** Stage **II** may involve the use of $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COCl}$ and methylamine.

- 28 Which of the following compounds are formed when $\text{CH}_3\text{CO}_2\text{CH}_2\text{NHCOCH}_3$ is heated under reflux with excess aqueous sodium hydroxide?

- A $\text{CH}_3\text{CO}_2\text{CH}_2\text{NH}_2$ and CH_3COOH
 B $\text{CH}_3\text{CO}_2\text{CH}_2\text{NH}_2$ and CH_3COONa
 C CH_3COOH and HOCH_2NH_2
 D CH_3COONa and HOCH_2NH_2

- 29 A section of nylon-6,6 is shown below.



Which monomers could form this polymer?

- A $\text{NH}_2(\text{CH}_2)_6\text{NH}_2$ and $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$
 B $\text{NH}_2(\text{CH}_2)_6\text{NHCO}_2\text{H}$ and $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$
 C $\text{NH}_2(\text{CH}_2)_6\text{NH}_2$ and $\text{HO}_2\text{C}(\text{CH}_2)_4\text{CO}_2\text{H}$
 D $\text{CH}_3(\text{CH}_2)_5\text{NH}_2$ and $\text{HO}_2\text{C}(\text{CH}_2)_4\text{CONH}_2$

- 30 LDPE is used to make plastic bags while HDPE is used to make milk jug.

Which of the following statements is true about these substances?

- 1 LDPE is flexible and HDPE is rigid.
 2 LDPE comprises of shorter branched polymer chains while HDPE comprises of longer linear polymer chains.
 3 LDPE and HDPE can be generated from propene using appropriate catalysts and reaction conditions.

- A 1, 2 and 3
 B 1 and 2 only
 C 1 only
 D 1 and 3 only

END OF PAPER